Multiobjective Optimization and Multiphysics Design of a 5 MW High-Speed IPMSM Used in FESS Based on NSGA-II

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***Abstract*—The flywheel energy storage system (FESS) has high-power density and fast response speed, hence it is widely used as a pulse power supply. For the high-speed permanent magnet synchronous machine (HSPMSM) used in high-power FESS, there are two technical problems that need to be solved, including huge centrifugal forces of the rotor at a high-speed operation region and large phase currents at a low-speed operation region. In this paper, a 5 MW high-speed interior PMSM (HSIPMSM) is designed for FESS. The design goal of this electric machine is to reduce the maximum phase current at the lowest speed as much as possible while ensuring the rotor mechanical strength. Based on the response surface (RS) model and non-dominated sorting genetic algorithm II (NSGA-II), the multiobjective design of the HSIPMSM is accomplished using 29 groups of multiphysical field finite element analysis (FEA) sampling experiments. The computational cost is effectively reduced. In addition, the constant power optimal trajectory and control switching speed of the HSIPMSM are analyzed and discussed, and the reason that the constant power flux weakening region of the designed HSIPMSM is not wide and the maximum torque per ampere (MTPA) mode is adopted at a low-speed operation region is further illustrated.**

***Index Terms*—Flux weakening, flywheel energy storage system, high-speed interior permanent magnet synchronous machine, multiobjective optimization, rotor mechanical strength.**

I. Introduction

GENERALLY, the flywheel energy storage system (FESS) has the features of high-power density, long service life, environmental protection, deep discharge, and fast response speed, such that it is well suited as a pulse power supply to complete energy output with hundreds of kilowatts or even megawatts in tens of seconds to minutes [1], [2]. To accomplish the energy conversion between mechanical energy and electrical energy, a high-speed electric machine should be used in FESS. Compared to other types of electric machines, the high-speed permanent magnet synchronous machine (HSPMSM) has many advantages, including high-power density, simple structure, and high efficiency. Therefore, an HSPMSM is usually more suitable as a motor/generator in the high-power FESS [3], [4]. Due to its simple structure, the surface PM rotor is often used in the HSPMSM, and some protective sleeves are employed to fix PMs in the shaft through the interference fit [5]. In addition, to avoid stress concentration and reduce eddy current loss, a PM pole is composed of multiple PMs [6]. To minimize air friction loss, the rotor usually rotates in vacuum magnetic levitation conditions. Unfortunately, the heat in the rotor produced by iron loss could be only dissipated via radiation, and the temperature of rotor would rise, even causing the irreversible demagnetization of PM [7], [8]. The design of HSPMSM used in FESS is a multiphysics comprehensive analysis process. Some factors that should be considered include the rotor mechanical strength and dynamics, electromagnetic field and charge-discharge control strategy, energy loss and temperature [9], [10].

In recent years, HSPMSMs with rated power of hundreds of kilowatts have emerged. With the increasing popularity of short-term high-power pulse power supply, the megawatt-level HSPMSM used in FESS has become popular research [11], [12]. As an FESS needs to discharge with rated power in the whole operation speed range, the inverter capacity of the FESS depends on the maximum phase current at the lowest speed. The maximum phase current of the high-power HSPMSM is very large, resulting in a significant increase in the capacity and cost of the FESS inverter [13], [14]. Compared with a surface-mounted PMSM, the saliency ratio and constant power speed range of the interior PMSM is larger and wider [15]. If the reluctance torque of the interior PMSM is effectively used, the maximum phase current of electric machine would be further reduced [16]. However, compared with a surface PM rotor, the stress concentration is more likely to occur inside the interior PM rotor [17]. As the volume and power level of the HSPMSM increase, the PM rotor with a larger diameter has to tolerate greater centrifugal forces [18]. Therefore, a comprehensive design of the megawatt-level high-speed interior PMSM (HSIPMSM) is imperative.

In the existing multiobjective designs of electric machines, the electromagnetic performance of electric machine has been optimized [19], [20]. However, the multiobjective optimization and multiphysics design of the high-power HSIPMSM is missing. In this paper, the main contribution of the work is

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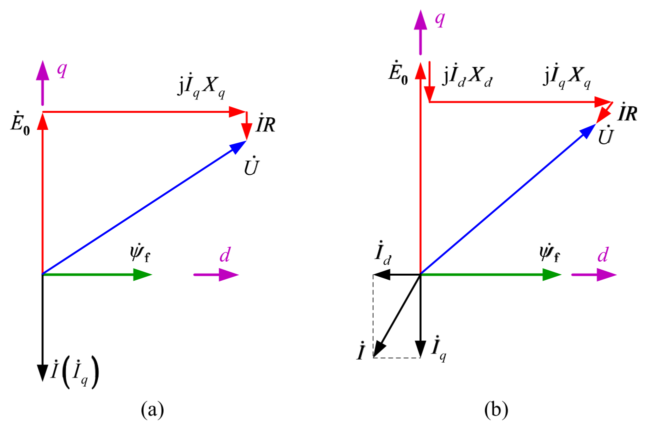

Fig. 1. Phasor diagrams of the HSIPMSM discharged at the highest speed under the motor reference convention. (a) Without the *d*-axis demagnetization current. (b) With the *d*-axis demagnetization current.

the effective multiobjective optimization design of a 5 MW HSIPMSM, which is realized through a low computational cost multiphysical field finite element analysis (FEA). While ensuring the rotor mechanical strength, the maximum phase current of the HSIPMSM is significantly reduced, and a smaller-capacity inverter can thus be used. The remaining sections are organized as follows. In Section II, the discharge operation constraints and optimal operation point are analyzed. In Section III, based on the response surface (RS) model and non-dominated sorting genetic algorithm II (NSGA-II), the optimal HSIPMSM is obtained, which has better mechanical performance, fewer PMs and smaller maximum phase current. In Section IV, the reason that the constant power flux weakening (FW) region of the HSIPMSM is not wide is discussed. Furthermore, the constant power optimal trajectory and control switching speed are further analyzed. In Section V, the performances of the electric machine are analyzed.

II. Constraints and Optimal Operation Point

Under the motor reference convention, the phasor diagrams of the HSIPMSM discharged at the highest speed are shown in Fig. 1. Fig. 1(a) shows the phasor diagram without the *d*-axis demagnetization current. Under this condition, the phase current is only the *q*-axis current. The phasor of the reactance voltage drop generated by *q*-axis current is orthogonal to the phasor of the line electromotive force (EMF) induced by PM flux linkage. The phasor sum of the EMF and impedance voltage drop is the phasor of the line voltage. The rated phase current is large due to the high-power level of the HSIPMSM, and the corresponding armature reaction is strong. To avoid that the line voltage amplitude exceeds the DC bus voltage during load operation, the EMF has to be small. The designed amplitude of the no-load line EMF at the highest speed could only be about 70% of the DC bus voltage, which leads to a very large phase current when the HSIPMSM discharges at the rated power and lowest speed. Fig. 1(b) shows the phasor diagram with the *d*-axis demagnetization current. Compared to Fig. 1(a), the PM flux linkage in Fig. 1(b) is larger, which makes the amplitude of the no-load line EMF at the highest speed close to 90% of the DC bus voltage. Due to the *d*-axis demagnetization current, the amplitude of the line voltage can be less than the DC bus voltage during load operation. Under rated power load operation, although the phase current with a large line EMF is larger than that with a small line EMF at the highest speed, the maximum phase current with a large line EMF is only 77.8% of that with a small line EMF at the lowest speed. Therefore, the PM flux linkage and EMF of the HSIPMSM are designed to be large; this is done to reduce the maximum phase current and the capacity of inverter. Next, the discharge operation constraints of the HSIPMSM and optimal operation point are analyzed.

A. Constraints

For an HSPMSM without the *d*-axis demagnetization current, there are four discharge operation constraints: speed, voltage, current and power constraint [21]. In this article, the *d*-axis demagnetization current is applied, such that the *d*-axis demagnetization current constraint needs to be considered.

1) Speed Constraint: The maximum storage energy E_s and maximum discharge energy E_d of FESS are

$$E_\mathrm{s} = \frac{1}{2} J\Omega_\mathrm{max}^2 \tag{1}$$

$$E_\mathrm{d} = \frac{1}{2} J\left(\Omega_\mathrm{max}^2 - \Omega_\mathrm{min}^2\right) \tag{2}$$

where J is the rotary inertia (kg·m^2). Ω_max and Ω_min are the maximum and minimum mechanical angular speed (rad/s).

The HSPMSM used in FESS is typically used for mechanical energy storage rather than speed regulation. The speed ratio is defined as the ratio of the highest speed to the lowest speed. If the speed ratio is set to 2, E_d would reach 75% of E_s, and the FESS could complete a deep discharge. Under this condition, the phase current at the lowest speed is 1.5 $\sim$ 2 times of that at the highest speed. If the speed ratio is set to 4, E_d would reach 93.75% of E_s. Although the energy released with a speed ratio of 4 is 18.75% more than that with a speed ratio of 2, only few benefits are obtained in terms of energy storage. However, when the speed ratio is 4, the maximum phase current at the lowest speed is close to 3 $\sim$ 4 times of that at the highest speed. The maximum phase current is too large, which would greatly increase the volume of electric machine and the capacity of inverter. Therefore, the speed ratio of high-power HSPMSM used in FESS is generally set to 2, and Ω_min is half of Ω_max.

$$\Omega_\mathrm{max} = 2\Omega_\mathrm{min} \tag{3}$$

The speed constraint equation is

$$\Omega_\mathrm{min} \le \Omega \le \Omega_\mathrm{max} \tag{4}$$

$$\Omega = \omega_\mathrm{e}/p_\mathrm{n} = 2\pi f/p_\mathrm{n} = 2\pi n/60 \tag{5}$$

where Ω and ω_e are the mechanical and electrical angular speed (rad/s). f is the electrical frequency (Hz). p_n is the number of pole pairs. n is the speed (r/min).

2) Voltage Constraint: In the synchronous rotating reference frame, the voltage equation at the steady-state condition could

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be simplified to

$$\begin{bmatrix} u_d \\ u_q \end{bmatrix} = \begin{bmatrix} R_\mathrm{s} & -\omega_\mathrm{e} L_q \\ \omega_\mathrm{e} L_d & R_\mathrm{s} \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_\mathrm{e} \psi_\mathrm{f} \end{bmatrix} \tag{6}$$

where u_d and u_q are the d-axis and q-axis voltage (V). i_d and i_q are the d-axis and q-axis current (A). L_d and L_q are the d-axis and q-axis inductance (H). R_s is the phase resistance (Ω). ψ_f is the PM flux linkage (Wb).

As the line voltage amplitude u_s shall not exceed DC bus voltage U_DC, the voltage constraint equation is

$$u_\mathrm{s}^2 = u_d^2 + u_q^2 \leq \left(\frac{U_\mathrm{DC}}{\sqrt{3}}\right)^2 \tag{7}$$

Ignoring the influence of phase resistance, we obtain

$$(\psi_\mathrm{f} + L_d i_d)^2 + L_q^2 i_q^2 \leq \frac{U_\mathrm{DC}^2}{3\omega_\mathrm{e}^2} \tag{8}$$

3) Current Constraint: During load operation, the phase current amplitude i_s shall not exceed the limit current i_lim provided by the inverter, and the current constraint equation is

$$i_\mathrm{s} = \sqrt{i_d^2 + i_q^2} \leq i_\mathrm{lim} \tag{9}$$

4) D-axis Demagnetization Current Constraint: During load operation, the d-axis demagnetization current is often applied. However, if the applied d-axis demagnetization current is excessive, the PMs may be demagnetized. Ignoring the magnetic saturation and cross-coupling effect, the d-axis demagnetization magnetomotive force (MMF) generated by the d-axis demagnetization current of the stator winding is completely used to offset the PM linkage; the d-axis demagnetization current constraint equation is approximately

$$i_d > i_{d\,\mathrm{lim}} \tag{10}$$

$$i_{d\,\mathrm{lim}} = \psi_\mathrm{f} / L_d \tag{11}$$

where $i_{d\,\mathrm{lim}}$ is the limit d-axis demagnetization current (A).

5) Power Constraint: Without regard to various losses, the mechanical power is approximately equal to the electromagnetic power, and the power constraint equation is

$$P_\mathrm{em} = T_\mathrm{em} \Omega \tag{12}$$

$$T_\mathrm{em} = \frac{6}{2} p_\mathrm{n} \left[\psi_\mathrm{f} i_q + (L_d - L_q) i_d i_q\right] \tag{13}$$

where P_em is the electromagnetic power (W). T_em is the electromagnetic torque (N·m).

From (13), the electromagnetic torque contains the permanent magnet torque and reluctance torque. If the reluctance torque produced by the difference between L_d and L_q is fully utilized, the maximum electromagnetic torque would be acquired with the minimum phase current.

In the motor reference convention, the operable region under the five constraints is inside the voltage limit (VL) ellipse determined by the DC bus voltage and current limit circle, on the right side of the d-axis demagnetization current limit line, and above the constant torque (CT) curve. The operable region meeting these constraints is the blue area in Fig. 2.

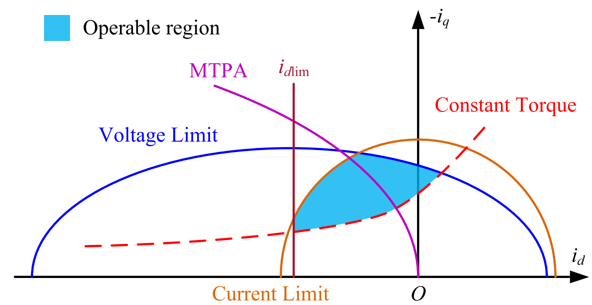

Fig. 2. Constraints and operable region of the HSIPMSM.

B. Optimal Operation Point

For the IPMSM used for speed regulation of the drive system, its constant power flux weakening region is wide. However, the constant power flux weakening region of the HSIPMSM used in FESS is not wide, which is caused by the large electromagnetic air gap of electric machine. There are three main reasons for the large electromagnetic air gap. First, because of the entire rotating system of the FESS in vacuum, the main energy loss of the FESS during long-term high-speed no-load standby operation is the stator iron loss. To reduce the stator iron loss and self-discharge rate of the FESS, the air-gap magnetic flux density of electric machine could not be designed too large, which is generally designed at 0.5 T $\sim$ 0.6 T; and the electromagnetic air gap is thus large. Second, due to the HSIPMSM with high-power level, its phase current is large and a strong armature reaction occurs under the rated power load. To avoid the serious magnetic saturation and reduce the stator iron loss during load operation, it is also necessary to design a large electromagnetic air gap. Third, to ensure the safety of high-speed PM rotor, a thick nonmagnetic carbon fiber sleeve is essential to be interference fitted at the outside of the entire rotor, which increases the electromagnetic air gap. As the constant power flux weakening region of the HSIPMSM is not wide, the maximum torque per ampere (MTPA) mode would be adopted at a low-speed operation region.

When the HSIPMSM discharges with a specific power at a certain speed, it is necessary to select the optimal operation point. In this article, the principle of minimum phase current is followed. In the case of a section CT curve of corresponding electromagnetic torque within the operable region, if the MTPA point (the intersection of MTPA curve and CT curve) is also within the operable region, the MTPA mode would be adopted and the electric machine would operate at this MTPA point. In contrast, the FW mode would be adopted and the electric machine would operate at the FW point (the intersection of VL curve and CT curve). The optimal operation point selection and constant power optimal trajectory of the HSIPMSM are shown in Fig. 3. The electric machine discharges with constant power from the highest speed n_1. As the MTPA point M_1 is not within the operable region, it operates at the FW point F_1. When the speed drops to n_2, the MTPA point M_2 is still not within the operable region, thus it operates at the FW point F_2. When the speed drops to n_3, the MTPA curve, VL curve and CT curve intersect

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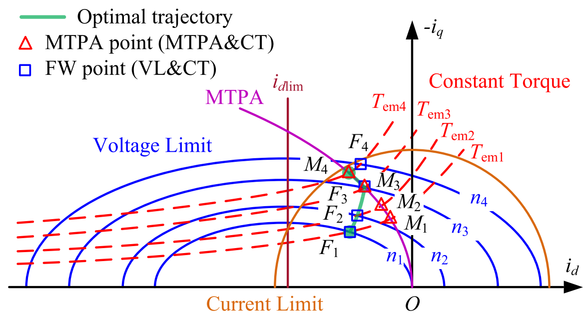

Fig. 3. The optimal operation point selection and constant power optimal trajectory of the HSIPMSM.

at the same point. That is to say, the M_3 point and F_3 point coincide. Therefore, the speed n_3 is the control switching speed of constant power discharge operation. After that, the MTPA mode can be adopted. When the speed drops to n_4, it operates at the MTPA point M_4. The optimal operation points of each speed are connected to obtain the constant power optimal trajectory. To summarize, when the speed is higher than the control switching speed, the FW mode is adopted, otherwise the MTPA mode is adopted. In the whole constant power discharge operation, with the decrease of speed, i_d firstly decreases and then increases, whereas i_s and i_q always increase.

III. Multiobjective Optimization of HSIPMSM

As different design variables have different effects on the performance of electric machine, there are often contradictions between them [22]. In this paper, the combination of RS model and NSGA-II algorithm is used for the multiobjective optimization of the interior PM rotor. The proposed optimization method includes the four steps: initial design, sensitivity analysis of design variable, RS model generation, and multiobjective optimization. The overall flowchart of optimization design of the HSIPMSM is shown in Fig. 4.

A. Initial Design

Compared to a general PMSM, the entire rotor dynamics analysis of the high-power HSIPMSM is particularly important. Hence, the initial electric machine design is based on design requirements and rotor dynamics. In the FESS, the rotor shaft and flywheel are made of superhigh strength alloy steel 35CrMnSiA. The outer diameter of the flywheel is 1290 mm, which is much larger than that of rotor. The linear speed of flywheel reaches 337.72 m/s at the highest speed of 5000 r/min, which is close to the sound speed. The system storage energy is 25 kWh. The cross section of the designed HSIPMSM is shown in Fig. 5. The main initial design parameters are listed in Table I. As the highest speed of the electric machine is 5000 r/min, the number of rotor poles is set to 6. Compared with the two-pole PMSM, although the electrical frequency and stator iron loss are increased, the span of stator winding and axial length of shaft are concurrently reduced. In addition, for the lower cost of inverter and better performance of electric machine, the 2Y six-phase winding is adopted, which could halve the current of phase winding and

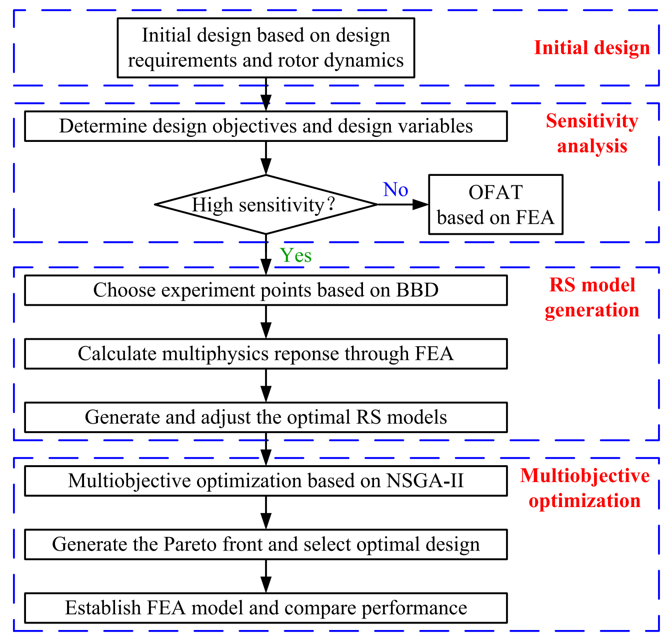

Fig. 4. Overall flowchart of optimization design of the HSIPMSM.

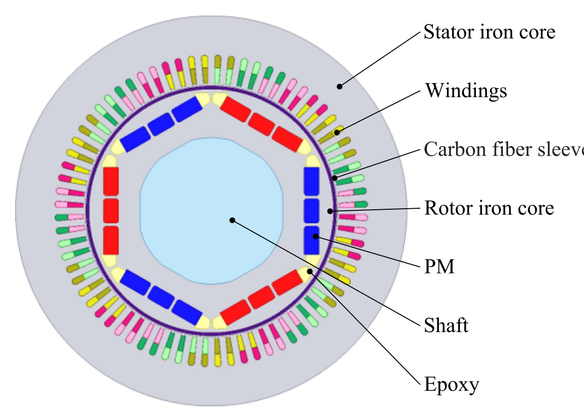

Fig. 5. Cross section of the designed 72 s/6 p HSIPMSM.

TABLE I
Main Initial Design Parameters of HSIPMSM

Parameter	Value	Parameter	Value
Rated power (MW)	5	Core length (mm)	555
Highest speed (r/min)	5000	Stator outer diameter (mm)	800
Lowest speed (r/min)	2500	Stator inner diameter (mm)	516
DC bus voltage (V)	1050	Rotor outer diameter (mm)	500
Phase/Slot/Pole number	6/72/6	Rotor inner diameter (mm)	300
Carbon fiber sleeve (mm)	6	Air gap length (mm)	2

remove the negative influence of some armature reaction MMFs [23].

To reduce centrifugal force and make full use of reluctance torque, the "I-shape" interior PM rotor is adopted and the width of two PMs on both sides is the same [24]. When the rotor of the HSIPMSM used in FESS is rotating in vacuum, the heat in the rotor is only dissipated via radiation, resulting in rotor heat dissipation difficulties. As the FESS is used as a high-power pulse power supply in important occasions, the safe

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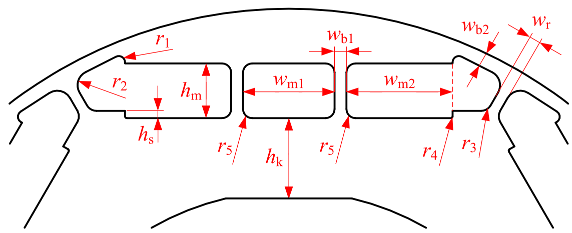

Fig. 6. Key design variables in one-pole interior PM rotor.

TABLE II
VALUE RANGE, INITIAL DESIGN AND OPTIMAL DESIGN VALUE OF DESIGN VARIABLE

Design variable	Value range (mm)	Initial design (mm)	Optimal design (mm)
Step height h_s	3-6	5	4
PM thickness h_m	26-34	31	30
Flat key height h_k	37-45	38	44
1st iron bridge width w_{b1}	4-8	7.5	6.5
2nd iron bridge width w_{b2}	4-8	7	7.2
Pole-to-pole rib width w_r	4-8	6.5	6
1st PM width w_{m1}	47-54	52	50
2nd PM width w_{m2}	55-62	57	58
1st fillet radius r_1	2-6	5	3.4
2nd fillet radius r_2	5-10	9	8
3rd fillet radius r_3	5-10	7	6
4th fillet radius r_4	1-3	2	1
5th fillet radius r_5	3-7	6	5

and reliable operation of the system is primarily considered. Considering the temperature stability of the PM, the slightly expensive N45UH is used in the designed HSIPMSM, whose limit operating temperature reaches 180 °C. In this design, other materials except the PM are the same as those in [25]. For the better mechanical strength, the interference fit is essential. Considering material properties and production technology, the interference value between the carbon fiber sleeve and rotor core is set to 0.25 mm.

B. Sensitivity Analysis of Design Variable

Based on reliable mechanical strength, the HSIPMSM is expected to have better electromagnetic performance. Therefore, the design objectives are the line EMF amplitude $E_{\max}$ and the maximum stress $\sigma_{\max}$ in the rotor core at the highest speed of 5000 r/min, and the maximum current $I_{\max}$ with rated power of 5 MW at the lowest speed of 2500 r/min. The multiobjective optimization function can be expressed as

$$\text{Function} : [\max E_{\max}, \min \sigma_{\max}, \min I_{\max}] \tag{14}$$

Fig. 6 displays the key design variables in one-pole interior PM rotor. The value range, initial design and optimal design value of design variable are listed in Table II. A larger fillet radius r_1 could reduce the maximum stress at this position, but the magnetic flux leakage between poles would be increased. In addition, the value range of r_1 is affected by the flat key height h_k, PM thickness h_m, and iron bridge width w_{b2}. The smaller h_k and larger h_m would result in greater centrifugal forces from the interior PMs and the rotor core outside PMs, and the stress in the area around the iron bridges with width of w_{b1} and w_{b2} would thus increase. The smaller the w_{b2}, the greater the stress in the area around the iron bridge with width of w_{b2}. For these reasons, the value range of r_1 is relatively small. For the fillet radius r_5, the larger r_5 could reduce the maximum stress at this position, but the magnetic flux leakage inside the PM pole would be increased and the utilization of the PMs would be reduced. Therefore, the value range of r_5 is also relatively small. For the fillet radiuses r_2, r_3, r_4, they have little effect on the magnetic flux leakage, and they are basically not affected by other variables. To reduce the stress at the fillet radiuses of r_2 and r_3, the value ranges of r_2 and r_3 are designed to be larger.

Inside the rotor core, the centrifugal forces from the interior PMs are mainly borne by the iron bridges with width of w_{b1} and w_{b2}. Due to the edging effect at the fillet of the iron bridge, the stress concentration is easy to occur in the rotor [26]. As the values of r_1 and r_5 are relatively small, the stress concentration can be easily caused. The maximum stress usually occurs at these two positions. Because of the protection of the carbon fiber sleeve, the stress in the area around the pole-to-pole rib with the width of w_r is small. In addition, the values of r_2 and r_3 are larger, which further avoids the stress concentration occurred at these two positions. Therefore, the stresses at the fillet radiuses of r_2 and r_3 are small. To simplify the optimization model and reduce the calculation, the stresses at the fillet radiuses of r_2 and r_3 are not included in the maximum stress $\sigma_{\max}$ function, which can be expressed as

$$\sigma_{\max} = \max(\sigma_{r1}, \sigma_{r5}) \tag{15}$$

If the number of design variables is large, the sample size of multiobjective optimization will be huge. For this reason, it is necessary to analyze the sensitivity of design variable and identify the impact of each variable on the performance of the electric machine. In this paper, the influence of some design variables on the design objectives is not monotonic, so the sensitivity coefficient S_{n_i} is taken at the absolute value, which can be expressed as

$$S_{n_i} = \left| \frac{\partial g}{\partial x_i} \bigg|_{\text{NOP}} \frac{x_i}{g} \right| \approx \left| \frac{\text{percentage change in } g}{\text{percentage change in } x_i} \right| = \left| \frac{\Delta g / g}{\Delta x_i / x_i} \right| \tag{16}$$

where g is the design objective and x_i is the design variable.

To evaluate the global influence of each design variable on three optimization objectives, the comprehensive sensitivity coefficient $G(n_i)$ and weight coefficient are introduced.

$$\begin{cases} G(n_i) = k_{E_{\max}} S_{E_{\max}} + k_{\sigma_{\max}} S_{\sigma_{\max}} + k_{I_{\max}} S_{I_{\max}} \\ k_{E_{\max}} + k_{\sigma_{\max}} + k_{I_{\max}} = 1 \end{cases} \tag{17}$$

where $S_{E_{\max}}$, $S_{\sigma_{\max}}$, and $S_{I_{\max}}$ are the sensitivity coefficients of $E_{\max}$, $\sigma_{\max}$, and $I_{\max}$. $k_{E_{\max}}$, $k_{\sigma_{\max}}$, and $k_{I_{\max}}$ are the corresponding weights. As the rotor mechanical strength is important, $k_{\sigma_{\max}}$ is set to 0.6. $k_{E_{\max}}$ and $k_{I_{\max}}$ are set to 0.2.

The comprehensive sensitivity analysis results are presented in Fig. 7. The comprehensive sensitivity coefficients of design variables h_k, h_m, w_{b1} and w_{b2} are the maximum, which are 0.386, 0.374, 0.175, and 0.166, respectively. For the other design variables, they are optimized by one factor at a time strategy (OFAT) based on FEA.

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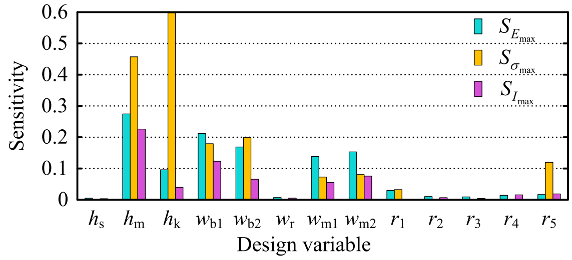

Fig. 7. Comprehensive sensitivity analysis of design variable.

C. RS Model Generation

Through the analysis of FEA sampling experimental results and the rational design of multiple regression equation, the RS model can establish an appropriate mathematical model between the design variable and objective. Among the common design of experiment (DoE) methods, the experiment times of the Box-Behnken design (BBD) method is less, and its sampling experimental value does not exceed the value range. Therefore, the BBD method is suitable to establish the RS model in this article, and only 29 groups of multiphysical field FEA sampling experiments are required to calculate. And then, various mathematical models are used to fit, and the second-order regression model is the most suitable, which can be expressed as

$$Y = \beta_0 + \sum_{i=1}^{m} \beta_i X_i + \sum_{i=1}^{m} \beta_{ii} X_i^2 + \sum_{i=1, i<j}^{m} \beta_{ij} X_i X_j \quad (18)$$

where Y is the predicted value of design objective. m is the number of design variables. β_0, β_i, β_{ii} and β_{ij} are the regression coefficients of different fitting terms. X_i and X_j represent two different design variables respectively.

Under the premise of accuracy, some nonsignificant terms are deleted to simplify RS models. The regression equation of the line EMF amplitude $E_{\max}$ is

$$\begin{aligned} E_{\max} = {} & 918 + 14.69h_m - 29.12w_{b1} - 24w_{b2} - 1.15h_k \\ & - 0.35h_m w_{b1} - 0.18h_m w_{b2} + 0.1h_m h_k + 0.2w_{b1}w_{b2} \\ & - 0.09w_{b1}h_k - 0.11h_m^2 + 0.58w_{b1}^2 + 0.14w_{b2}^2 + 0.02h_k^2 \end{aligned} \quad (19)$$

The regression equation of the stress σ_{r1} at the fillet r_1 is

$$\begin{aligned} \sigma_{r1} = {} & 1465.75 - 42.05h_m - 83.61w_{b1} - 20.67w_{b2} - 3h_k \\ & + 3.06w_{b1}w_{b2} + 0.724h_m^2 + 5.042w_{b1}^2 \end{aligned} \quad (20)$$

The regression equation of the stress σ_{r5} at the fillet r_5 is

$$\begin{aligned} \sigma_{r5} = {} & 1679.92 - 16.86h_m - 73.63w_{b1} - 53.05w_{b2} \\ & - 10.12h_k + 1.471h_m w_{b2} + 4.713w_{b1}^2 \end{aligned} \quad (21)$$

The regression equation of the maximum current $I_{\max}$ is

$$\begin{aligned} I_{\max} = {} & 5026.3 - 132.1h_m - 5.1w_{b1} + 85.2w_{b2} \\ & + 20.8h_k - 0.7h_m w_{b2} + 0.22h_m h_k + 1.16w_{b1}w_{b2} \\ & + 1.23w_{b1}h_k - 0.85w_{b2}h_k + 1.667h_m^2 \\ & + 2.76w_{b1}^2 + 0.49w_{b2}^2 - 0.333h_k^2 \end{aligned} \quad (22)$$

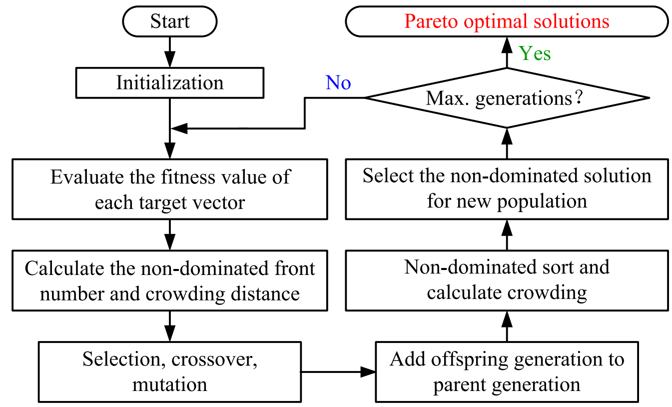

Fig. 8. Flowchart of NSGA-II algorithm.

For the benefit of correlation and accuracy of the RS model, the determination coefficient R^2 and root mean square error (RMSE) are used for significance test and variance analysis. R^2 is defined as

$$R^2 = 1 - \left(\sum_{i=1}^{n} (y_i - f_i)^2 \Big/ \sum_{i=1}^{n} (y_i - \bar{y})^2 \right) \quad (23)$$

where n is the number of sampling experiments, which is 29 in this article. y_i is the desired value from FEA. f_i is the predicted value by RS model. $\bar{y}$ is the average value of y_i.

Based on statistic theory, the better the model fits the data, the closer the R^2 value is to 1. In this article, the R^2 values of four RS models are greater than 0.99, indicating that they have good correlation and precision. The better the model stability is, the smaller the RMSE is; this is described as

$$\mathrm{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - f_i)^2} \quad (24)$$

After analysis, the RMSE values of four RS models are all less than 1%, so they have high accuracy and good suitability.

D. Multiobjective Optimization Based on NSGA-II

Considering the various performances of the HSIPMSM, the multiobjective optimization algorithm is adopted to analyze the RS models. As the non-dominated sorting and crowding distance are introduced to measure the distribution of solutions in the NSGA-II algorithm, the convergence of the solution set is better [27]. In this article, the NSGA-II algorithm is used to solve the multiobjective optimization design of the HSIPMSM, and the corresponding flowchart is shown in Fig. 8. Moreover, the population number is 2000 and the maximum iteration generation is 200. After the multiobjective optimization, the optimal design is selected from a set of optimal solutions (Pareto front). For the safety of the interior PM rotor, $\sigma_{\max}$ is expected to be as small as possible. Considering the actual engineering requirements, $\sigma_{\max}$ shall not exceed 400 MPa. And then, the optimal design is obtained with the line EMF amplitude of 926 V, the maximum stress of 387.8 MPa, and the maximum current of

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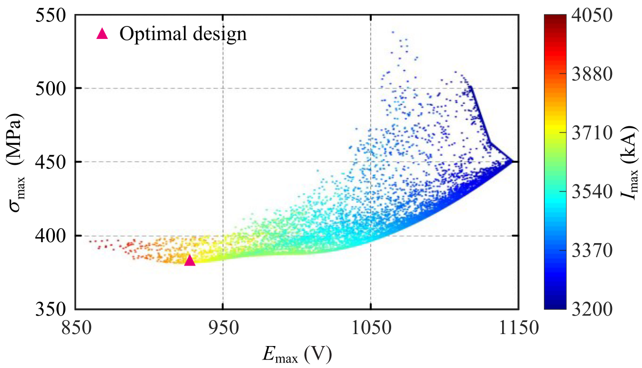

Fig. 9. Multiobjective optimization scatter plot and optimal design.

TABLE III
PERFORMANCE COMPARISON

Item	Initial design (FEA)	Optimal design (FEA)	Optimal design (RS model)
$E_{\max}$ (V)	888.5	926.5	926.0
$\sigma_{\max}$ (MPa)	432.6	387.0	387.8
$I_{\max}$ (A)	3867.1	3826.6	3828.0

3828 A. The multiobjective optimization scatter plot and optimal design are shown in Fig. 9.

The performance comparison between the initial design and optimal design is shown in Table III. The performance of the optimal electric machine based on RS model is basically same as that calculated by FEA, indicating the accuracy of the proposed multiobjective optimization method. From Table II and Table III, although the fewer PMs are used in the optimal HSIPMSM, its performance is better than the initial electric machine. The maximum phase current and capacity of inverter are concurrently reduced, and the high-speed operation safety is ensured. The locally optimal electric machine is achieved.

For four foremost design variables in this article, only 29 groups of FEA sampling experiments are calculated to establish the RS model. And then, the NSGA-II algorithm is combined with the RS model to achieve the multiobjective optimization. In this work, the system of the computer is Inter(R) Core(TM) i7-8750H CPU with 16-GB-RAM. The computation time for 1 group of multiphysical field FEA simulation is about 0.2 h, and the computation time of multiobjective optimization based on the NSGA-II algorithm is about 0.1 h, thus the total computation time of RS model and NSGA-II algorithm is 5.9 h. However, if a conventional multiobjective optimization is used, such as differential evolution or genetic algorithm directly combined with FEA, the more computation would be required to execute [28]. When the population number and the maximum iteration generation are set to 100 and 50 respectively, there are 5000 candidate design evaluations, and corresponding total calculation time is 1000 h. The comparison of computation time between these two methods is summarized in Table IV. With limited computational power, the combination of RS model and NSGA-II algorithm can greatly reduce the computation time of multiphysical field FEA simulation and multiobjective optimization.

TABLE IV
COMPARISON OF COMPUTATION TIME

Method	Time of FEA (h)	Time of optimization (h)	Total computation time (h)
RS model and NSGA-II	5.8 (29 groups)	0.1	5.9
Conventional optimization	1000 (5000 groups)	-	1000

IV. CONSTANT POWER OPTIMAL TRAJECTORY AND CONTROL SWITCHING SPEED

In the designed HSIPMSM, the thickness of the physical air gap and carbon fiber sleeve are 2 mm and 6 mm respectively. Hence, the electromagnetic air gap is equivalent to 8 mm, and the corresponding air-gap reluctance is large. Although the difference between d-axis inductance and q-axis inductance of the designed "I-shape" interior PM rotor is huge, the saliency ratio of the HSIPMSM is small due to the large air-gap reluctance. When the magnetic circuit is unsaturated during no-load operation, the unsaturated saliency ratio is 1.85. When the HSIPMSM discharges at the rated power and lowest speed, the magnetic circuit is saturated. Under this condition, the saliency ratio is only 1.37. The small saliency ratio makes the constant power flux weakening region of the designed HSIPMSM not wide, and the MTPA mode could be adopted at a low-speed operation region. In this section, the reason that the constant power flux weakening region of the HSIPMSM is not wide is analyzed. In addition, the constant power optimal trajectory and control switching speed of the HSIPMSM are further analyzed and discussed to illustrate the operation effect in the whole operation speed range.

The HSIPMSM used in FESS should act as both a motor and a generator to charge and discharge reversibly, and the analysis in this section is discussed under the generating mode. As the motor reference convention is adopted, the values of i_q are negative. As mentioned in the power constraint in Section II, without regard to various losses, the mechanical power is approximately equal to the electromagnetic power. Under this assumption, when the HSIPMSM operates at the same power and speed in motoring and generating modes, the i_d is the same; the values of i_q in both modes are also the same, only the phases of i_q are opposite. Therefore, analytical results in this section are applicable to the motoring mode, only the signs of i_q need to be changed to positive.

A. Constant Voltage and Constant Torque Distribution

For the discharge reliability, the effective value of limit current $i_{\lim}$ is set to 4000 A. The effective value range of i_d and i_q are all -3600 A $\sim$ 0 A, and 60 A is set as the step size. When the HSIPMSM rotates at 5000 r/min, the distribution of line voltage amplitude $U_{\max}$ and constant voltage (CV) curves are obtained by FEA, as shown in Fig. 10. $U_{\max}$ is mainly affected by i_q,

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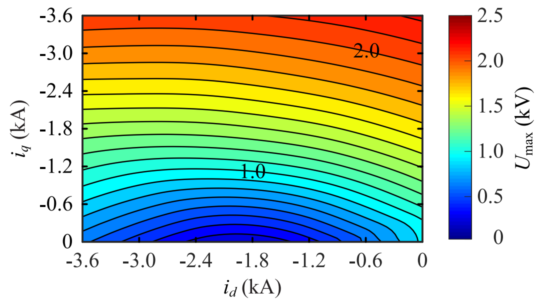

Fig. 10. The distribution of line voltage amplitude and CV curves at 5000 r/min.

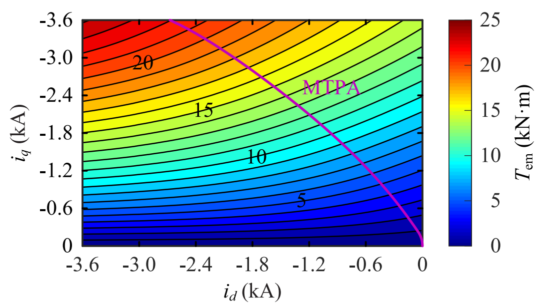

Fig. 11. The distribution of electromagnetic torque, CT curves and MTPA curve at 5000 r/min.

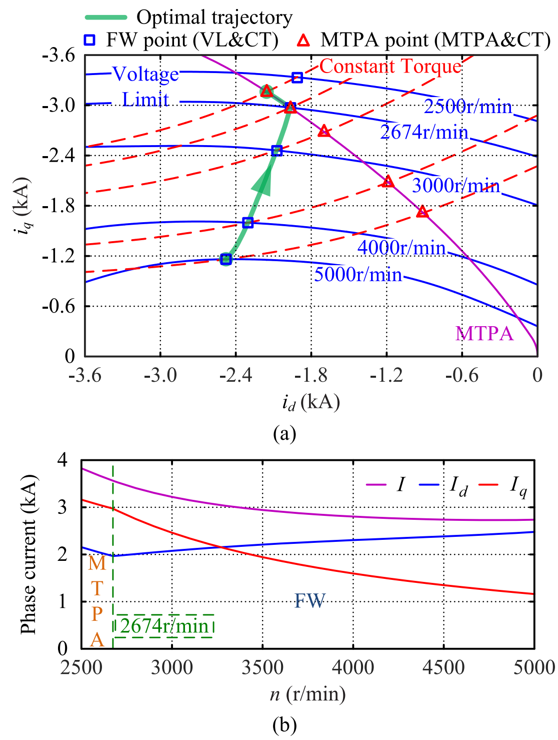

Fig. 12. Rated power 5 MW discharge operation. (a) The constant power optimal trajectory. (b) Corresponding phase current change.

and $U_{\max}$ increases with the increase of i_q. i_d also has a certain influence on $U_{\max}$. As i_d is *d*-axis demagnetization current, with the increase of i_d, $U_{\max}$ firstly decreases and then increases. In addition, the saturation and cross-coupling effect are evident due to the armature reaction. The CV curve is not a standard CV ellipse and the left side of the elliptical short axis is slightly higher than the right side. For the same reason, the center point of CV ellipse shifts with the increase of $U_{\max}$. The distribution of electromagnetic torque $T_{\rm em}$, CT curves and MTPA curve are presented in Fig. 11. $T_{\rm em}$ is mainly affected by i_q. i_d also has a certain influence on $T_{\rm em}$. When i_d is large, $T_{\rm em}$ increases rapidly with the increase of i_q. The reason is that according to (13), with the same i_q, a larger i_d can produce a larger reluctance torque.

B. Constant Power Optimal Trajectory

In the designed interior PM rotor, the PMs are not directly exposed to the electromagnetic air gap. The rotor core between the outside of the PMs and the outer diameter of the rotor provides a bypass path for the PM flux and armature reaction *d*-axis demagnetization flux, which can effectively weaken the effect of the *d*-axis demagnetization MMF on the PMs. For this reason, the limit *d*-axis demagnetization current in (11) basically has no effect on the designed interior PM rotor, and the *d*-axis demagnetization current constraint is thus not considered in the operation point selection of the HSIPMSM. For the FESS, the DC bus voltage $U_{\rm DC}$ is 1050 V. As the voltage drops of stator end-winding leakage inductance and phase resistance are not considered in the FEA simulation model, $U_{\rm DC}$ should be set lower in the FEA model, which is set to 1000 V. When i_d and i_q are determined, the main magnetic circuit characteristic of electric machine is determined. Under this condition, $T_{\rm em}$ is constant and $U_{\max}$ is only proportional to speed, and the VL curve at a certain speed can be obtained from Fig. 10. The MTPA curve and CT curve relative to a specific discharge power at this speed can be obtained from Fig. 11. Base on the theoretical analysis in Section II, the constant power optimal trajectory is obtained.

Under rated power 5 MW discharge operation, the constant power optimal trajectory and corresponding phase current change are presented in Fig. 12. From Fig. 12(a), the FW mode is adopted from 5000 r/min to 2674 r/min, and the MTPA mode is adopted from 2674 r/min to 2500 r/min. Under this condition, 2674 r/min is the control switching speed. From Fig. 12(b), with the decrease of speed, $i_{\rm s}$ and i_q always increase; i_d firstly decreases and then increases, and this change occurs at the control switching speed. In addition, the minimum phase current is 2737.1 A at 5000 r/min, and the maximum phase current is 3826.6 A at 2500 r/min. The ratio of maximum and minimum phase current is only 1.4, less than the speed ratio of 2. Although the minimum phase current at 5000 r/min is increased due to the *d*-axis demagnetization current applied, the maximum phase current at 2500 r/min is dramatically reduced. Thus, a smaller-capacity inverter can be used in the FESS.

In practice, the HSIPMSM needs to discharge according to the power requirement of system. Under half power 2.5 MW discharge operation, the constant power optimal trajectory and corresponding phase current change are shown in Fig. 13. Under this condition, The maximum and minimum phase current are 1941.9 A and 1116.1 A, respectively. Moreover, the control

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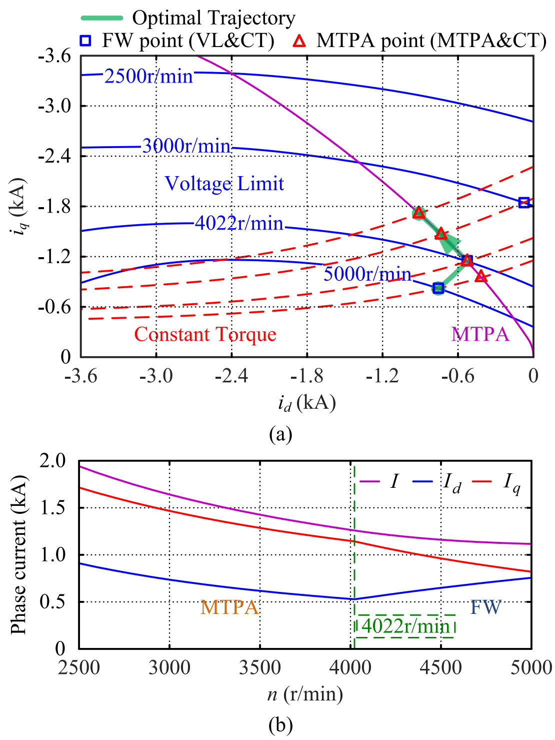

Fig. 13. Half power 2.5 MW discharge operation. (a) The constant power optimal trajectory. (b) Corresponding phase current change.

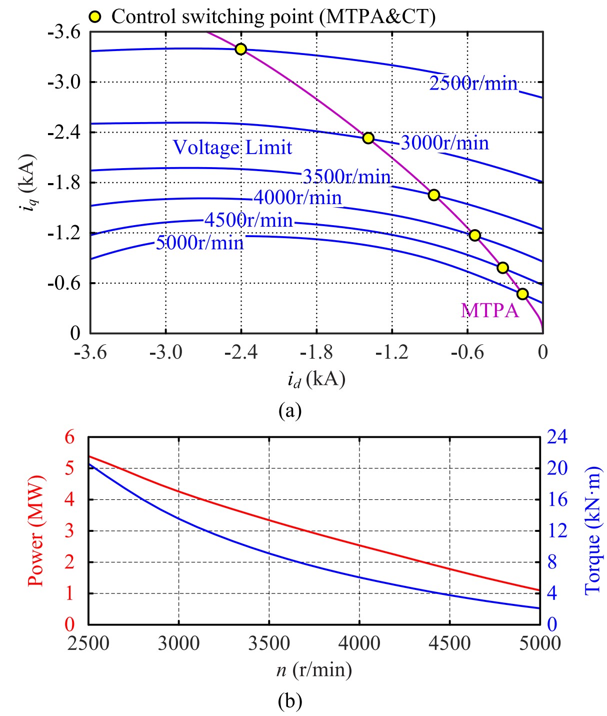

Fig. 14. (a) The control switching points at different speeds (the intersections of CT curves and MTPA curve). (b) The electromagnetic torque and power of control switching point in the whole operation speed range.

TABLE V
CONTROL SWITCHING SPEED AND MAXIMUM/MINIMUM CURRENT

Discharge power P (MW)	Switching speed n_s (r/min)	Maximum current I_{max} (A)	Minimum current I_{min} (A)
5.0	2674	3826.6	2737.1
4.0	3134	3042.9	1885.8
3.0	3704	2303.3	1348.1
2.5	4022	1941.9	1116.1
2.0	4354	1581.4	893.0
1.0	-	850.1	451.8

switching speed is 4022 r/min. As shown in Figs. 12 and 13, the control switching speed increases from 2674 r/min to 4022 r/min, and the MTPA mode is thus adopted earlier. The reason is that when the HSIPMSM discharges with half power of 2.5 MW, the armature reaction is weak and the required d-axis demagnetization current is small. The control switching speed and maximum/minimum phase current with different discharge powers are summarized in Table V. With the decrease of discharge power, the control switching speed gradually increases. When the discharge power is 1 MW, the MTPA mode can be directly adopted at 5000 r/min.

C. Control Switching Speed

To obtain the control switching points at different speeds, the CV curves from Fig. 10 are respectively intersected with the MTPA curve from Fig. 11, as shown in Fig. 14(a). The corresponding electromagnetic torque and power of control switching point in the whole operation speed range are presented in Fig. 14(b). The speed corresponding to a specific discharge power is the control switch speed of this discharge power. With the increase of discharge power, the control switching speed decreases. Similarly, the discharge power corresponding to a certain speed is the control switching power of this speed. With the increase of speed, the control switching power decreases. When the discharge power is greater than the control switching power at this speed, the FW mode is adopted, otherwise the MTPA mode is adopted. As the control switching power at 2500 r/min is 5.39 MW, the FW mode is adopted at a low-speed operation region when the HSIPMSM discharges with rated power of 5 MW.

V. TOTAL PERFORMANCE AND NO-LOAD EXPERIMENT

A. Mechanical Strength Analysis

When the FESS is not required to charge or discharge, the HSIPMSM would stand by at the highest speed of 5000 r/min. In this situation, the maximum centrifugal force would happen in the interior PM rotor, and the corresponding mechanical strength analysis result is displayed in Fig. 15. The stress of the rotor core is mainly caused by the centrifugal forces from the interior PMs and the rotor core outside PMs. As the entire rotor of the designed HSIPMSM is protected by a carbon fiber sleeve, the deformations of the PMs and rotor core are small, and the PMs are subjected to compressive stress. Inside the rotor core, the maximum deformation is occurred at the rotor core between the outside of the PMs and the outer diameter of the rotor, and the centrifugal forces are mainly borne by the iron bridges with width of w_{b1} and w_{b2}. As the entire rotor is protected, the stress in the area around the pole-to-pole rib with the width of w_r is

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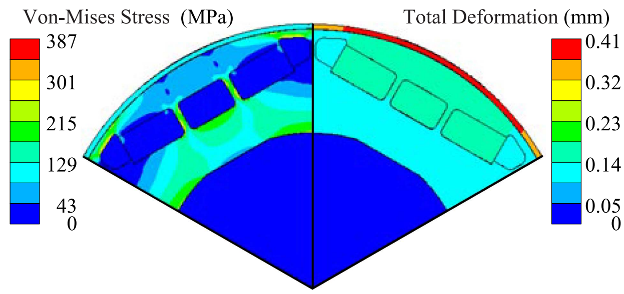

Fig. 15. Mechanical strength analysis of the "I-shape" interior PM rotor at the highest speed of 5000 r/min.

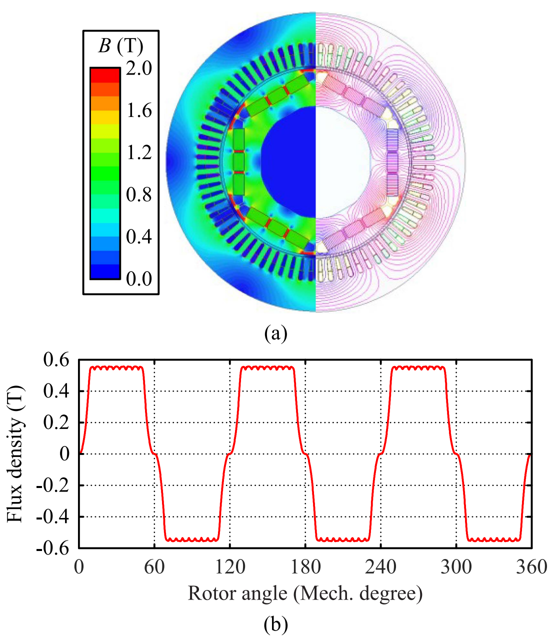

Fig. 16. No-load electromagnetic field analysis: (a) Color plot of magnetic flux density and whole flux lines. (b) Air-gap magnetic flux density waveform.

small. The maximum stress in the rotor core is occurred at the fillet radius r_1, which is 387 MPa. The value is less than the material tensile strength of 450 MPa. In addition, the maximum deformation in the carbon fiber sleeve is occurred at the section of the sleeve corresponding to the radial direction of the interior PMs, which is 0.41 mm.

B. Electromagnetic Field Analysis

The analytical results of the no-load electromagnetic field of electric machine are shown in Fig. 16. As the electromagnetic air gap and corresponding air-gap reluctance are large, the air-gap magnetic flux density is small. From Fig. 16(a), the no-load maximal flux density at the stator tooth is 0.86 T, and the average flux density at the stator yoke is 0.69 T. Although the no-load magnetic flux density is low, this design meets the low stator iron loss and low self-discharge rate during long-term no-load standby operation. From Fig. 16(b), the no-load air-gap magnetic flux density waveform is like a flat top wave, and the availability of iron core and power density of electric machine are thus improved. After the Fourier series expansion of flux density, the

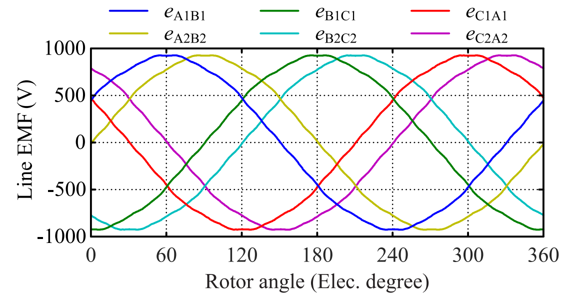

Fig. 17. No-load line EMF waveform of 2Y six-phase winding at 5000 r/min.

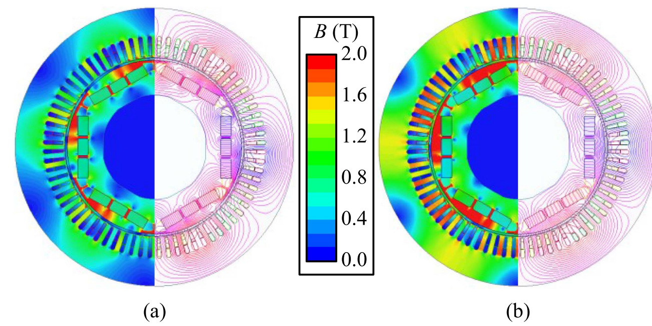

Fig. 18. Color plot of magnetic flux density and whole flux lines with rated power of 5 MW. (a) 5000 r/min with the maximum d-axis demagnetization current. (b) 2500 r/min with the maximum phase current.

amplitude of fundamental component is 0.66 T. Even though the amplitude of the 3rd flux density harmonic component is 0.13 T, the 2Y six-phase winding makes the 3rd harmonic does not produce the corresponding line EMF. In addition, the amplitudes of the 5th and 7th flux density harmonic components are only 0.01 T and 0.05 T respectively. In the designed HSIPMSM, the coil pitch is 10, and the winding coefficients of 5th and 7th harmonic components are thus 0.2053 and 0.1576, which can effectively restrain the negative influence of them. For these reasons, the line EMF waveform has better sine quality. The no-load line EMF waveform of 2Y six-phase winding at 5000 r/min is displayed in Fig. 17, and the line EMF waveform is close to a sine wave. After the Fourier series expansion, the fundamental line EMF component reaches 920.1 V. Among other EMF harmonic components, the amplitude of 7th EMF harmonic component is the maximum, which is only 11.8 V.

When the HSIPMSM discharges at the rated power and highest speed of 5000 r/min, the maximum d-axis demagnetization current of 2478 A is applied to ensure $U_{\max}$ less than U_{DC}. Under this condition, the color plot of magnetic flux density and whole flux lines are shown in Fig. 18(a). The magnetic saturation degree of stator core is low, such that the stator iron loss is not high. When the HSIPMSM discharges at the lowest speed of 2500 r/min, the phase current of 3826.6 A is maximum, and the d-axis demagnetization current of 2154 A is applied to make the HSIPMSM operate at the MTPA point. The corresponding electromagnetic field analytical results are shown in Fig. 18(b). In this condition, the maximal flux density at the tooth reaches 1.87 T. Fortunately, the speed is the lowest, so the stator iron loss is also acceptable. In addition, the PMs are not demagnetized in

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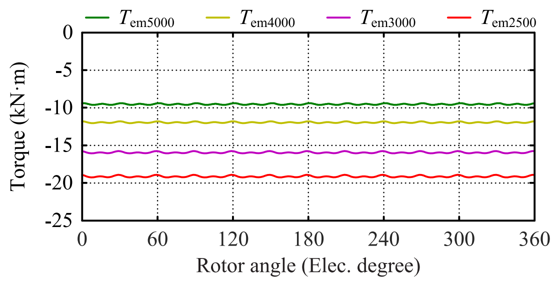

Fig. 19. Electromagnetic torques under rated power 5 MW discharge operation at different speeds.

TABLE VI
ELECTROMAGNETIC TORQUE UNDER RATED POWER 5 MW DISCHARGE OPERATION

Torque performance	5000 r/min	4000 r/min	3000 r/min	2500 r/min
T_{em} (kN·m)	-9.55	-11.94	-15.92	-19.10
ΔT (%)	2.24	2.09	1.90	1.81

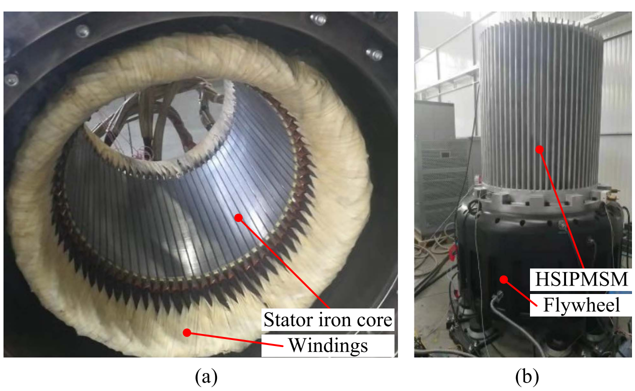

Fig. 20. (a) The entire stator structure with 72 slots six-phase winding. (b) The overall prototype of 5 MW FESS.

both extreme conditions. The reliable discharge operation of the electric machine can be guaranteed.

Under rated power discharge operation, the electromagnetic torque waveforms at different speeds are presented in Fig. 19, and the corresponding analytical results are summarized in Table VI. It can be observed that the torque ripple is small in the whole operation speed range. In addition, the peak value of cogging torque is only 2.5 N·m, which is small. Thus, the electric machine has great torque performance.

C. No-Load Experiment

The entire stator structure of the HSIPMSM and the overall prototype of 5 MW FESS are displayed in Fig. 20. After the prototype manufactured, the no-load experiment is carried out. A small-capacity six-phase inverter is used to supply power, and the electric machine as a motor drives the flywheel start and gradually accelerate. When the speed is higher than 5000 r/min, the inverter stops power supply and an ondograph is used to record the no-load line EMF waveform at 5004.5 r/min, as shown in Fig. 21(a). The waveform comparison between experiment

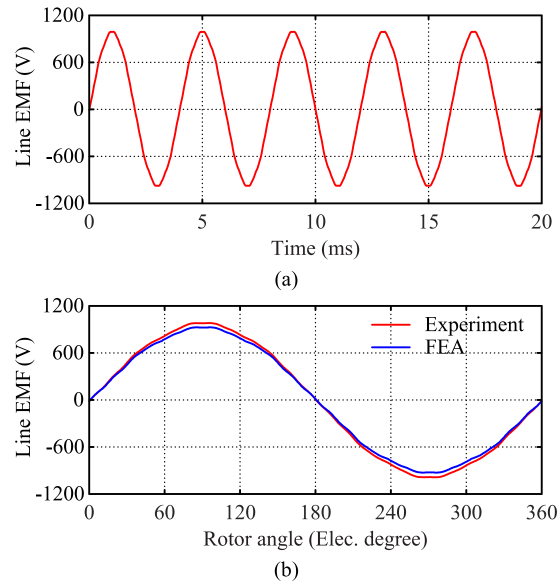

Fig. 21. (a) Measured waveform of no-load line EMF at 5004.5 r/min. (b) Waveform comparison between experiment and FEA simulation.

and FEA simulation is presented in Fig. 21(b). The shapes of them are basically similar, and their amplitudes are 984 V and 927.3 V at 5004.5 r/min. The EMF amplitude of experiment is 6.1% larger than that of FEA simulation. In practice, the FESS is required to charge and discharge. And then, the rotor iron loss would gradually increase the temperature of the interior PM rotor operating in vacuum. Furthermore, the higher rotor temperature may lead to irreversible demagnetization and temperature-dependent reversible demagnetization of PMs. In this situation, the amplitude of the line EMF would be basically consistent with the design value.

VI. CONCLUSION

In This article, the Multiobjective Optimization Design of a 5 MW HSIPMSM used in FESS is nicely achieved based on the combination of RS model and NSGA-II algorithm. While ensuring the rotor mechanical strength, the maximum phase current at the lowest speed is effectively reduced, thus a smaller-capacity inverter can be used. There are some conclusions drawn.

1) After the sensitivity analysis, the foremost design variables with maximum comprehensive sensitivity coefficients are the flat key height h_k, PM thickness h_m, iron bridge width w_{b1} and w_{b2}. After the multiobjective optimization, the optimal HSIPMSM has better mechanical performance, fewer PMs and smaller maximum phase current.
2) Although the difference between d-axis inductance and q-axis inductance of the designed "I-shape" interior PM rotor is huge, the saliency ratio of the HSIPMSM is small due to the large air-gap reluctance, which results in the constant power flux weakening region of the designed HSIPMSM not wide. For this reason, the MTPA mode can be adopted at a low-speed operation region. When the HSIPMSM discharges with rated power of 5 MW, the

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control switching speed is 2674 r/min, and the speed range of the MTPA mode is 2674 r/min $\sim$ 2500 r/min.

3) With the decrease of discharge power, the corresponding control switching speed increases. When the discharge power is 1 MW, the control switching speed is larger than 5000 r/min. Under this condition, the MTPA mode is adopted in the whole operation speed range.

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